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c. displacement and acceleration is π radian or 180°.

Solution

a. The displacement of the particle executing simple harmonic motion
$y = A \sin\omega t$

Velocity of the particle is

$$v = A\,\omega\cos\omega t = A\,\omega\sin\left(\omega t + \frac{\pi}{2}\right)$$

The phase difference between displacement and velocity is $\frac{\pi}{2}$.

b. The velocity of the particle is

$$v = A\,\omega\cos\omega t$$

Acceleration of the particle is

$$a = -A\omega^2\sin\omega t = A\omega^2\cos\left(\omega t + \frac{\pi}{2}\right)$$

The phase difference between velocity and acceleration is $\frac{\pi}{2}$.

c. The displacement of the particle is
$y = A \sin\omega t$

Acceleration of the particle is

$a = -A\,\omega^2 \sin\omega t = A\,\omega^2 \sin(\omega t + \pi)$

The phase difference between displacement and acceleration is π.

10.3 ANGULAR SIMPLE HARMONIC MOTION

10.3.1 Time period and frequency of angular SHM

When a body is allowed to rotate freely about a given axis then the oscillation is known as the angular oscillation. The point

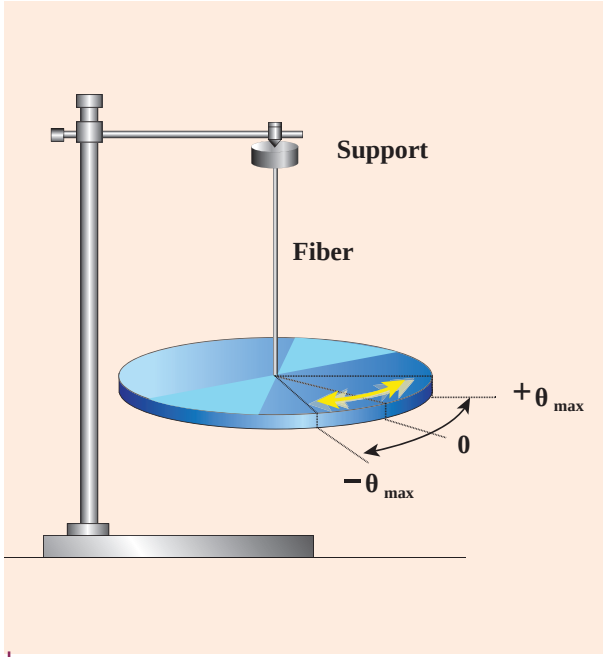


Figure 10.12 A body (disc) allowed to rotate freely about an axis

at which the resultant torque acting on the body is taken to be zero is called mean position. If the body is displaced from the mean position, then the resultant torque acts such that it is proportional to the angular displacement and this torque has a tendency to bring the body towards the mean position. (Note: Torque is explained in unit 5)
Let $\vec{\theta}$ be the angular displacement of the body and the resultant torque $\vec{\tau}$ acting on the body is

$$\vec{\tau} \propto \vec{\theta} \qquad (10.15)$$

$$\vec{\tau} = -\kappa\vec{\theta} \qquad (10.16)$$

κ is the restoring torsion constant, which is torque per unit angular displacement. If I is the moment of inertia of the body and $\vec{\alpha}$ is the angular acceleration then

$$\vec{\tau} = I\vec{\alpha} = -\kappa\vec{\theta}$$